

Shiva Balan K

LinkedIn: [linkedin.com/in/shiva-balan](https://www.linkedin.com/in/shiva-balan)

GitHub: github.com/shivabalank

Portfolio: shivabalan-portfolio.vercel.app/

Email: laxmi60097@gmail.com

Mobile: +91-9902133916

SKILL SUMMARY

- **Languages:** C, Python, SQL, JavaScript
- **Web Technologies:** HTML, CSS, Tailwind, JavaScript, React, Streamlit, Bootstrap, Kotlin
- **AI & ML:** Machine learning, Deep learning, LLM, RAG, Hugging Face Transformers, TensorFlow, LSTM, CNN, RNN, Scikit-learn, Prompt Engineering
- **Databases & Vector Stores:** MySQL, MongoDB, Google Firebase, Qdrant/FAISS
- **Data Analysis:** NumPy, Pandas Matplotlib/Seaborn, Excel
- **Tools:** Git, GitHub, VS Code, Android Studio, Google colab

EXPERIENCE

AiRobosoft – Full Stack Development Intern

Jun 2025 – Oct 2025

- Developed responsive web app components using HTML, CSS, JS enhancing user experience across devices.
- Integrated frontend interfaces with backend functionalities and databases to support dynamic application features.
- Participated in testing, debugging, and optimizing web applications, improving performance and reliability.

MindMatrix –Android App Development Using GenAI Intern

Sep 2025 – Jan 2026

- Built Android applications using Kotlin, following modern mobile development practices
- Leveraged Generative AI technologies to develop intelligent features and improve application functionality.
- Conducted testing, debugging, and performance optimization to deliver reliable and efficient mobile applications.

PROJECTS

RAG Based Document Assistant

JUN 2026 – Aug 2026

- Built a backend RAG (Retrieval-Augmented Generation) pipeline in Python using Llama-2-7B-Chat, Sentence-Transformers, and Qdrant to answer natural-language questions over PDF/DOCX documents with source-cited, hallucination-controlled responses.
- Implemented resumable document ingestion (checkpointed embedding/indexing), configurable chunking/retrieval/reranking, and a modular architecture designed for FastAPI deployment

DeepFake Detection using Efficient-B4

Jan 2026 – Apr 2026

- Developed a full-stack deepfake detection system using an EfficientNet-B4 CNN classifier with MTCNN-based face detection to identify AI-generated facial manipulations in images and video.
- Implemented Grad-CAM explainability to visualize the facial regions influencing each prediction. Built a Flask REST API backend and a React/TypeScript frontend, containerized with Docker and deployed on Google Cloud Run and Vercel.
- Achieved a test-set accuracy of **72%** with an ROC-AUC of **80%**, validated via recall, and F1-score metrics.

Employee Tracker Application Using Kotlin

Nov 2025 – Jan 2026

- Developed an Employee Tracker Android application using Kotlin with separate role-based authentication for HR and employees using Firebase Authentication.
- Integrated Firebase Realtime Database/Firestore for secure data storage, user management, and seamless synchronization of employee attendance data.

EDUCATION

Visvesvaraya Technological University

KARNATAKA, INDIA

Bachelor of Engineering – Computer Science and Engineering (AI&ML); CGPA: 7.8

Nov 2022 - June 2026

CERTIFICATES

- Oracle Cloud Infrastructure 2025 Certified Data Science Professional by Oracle Corporation
- Complete Web development Course by Hitesh Choudhary, Udemy